

UNILIS - Unified Learning and Innovation Lab Information System

Comprehensive Software Documentation & Feature Inventory

Version: **2.1.0**

Last Updated: **2026-07-28**

Total Features Documented: **95**

Comprehensive feature inventory for the UNILIS LMS system including Traditional LMS, SmartLab (SmartLab), Live Engagement, Teams, Chat, Meeting Server, ICLM (Self-Paced & Public Courses), and Public Learning (Learn) modules.

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1. Executive Summary

Comprehensive feature inventory for the UNILIS LMS system including Traditional LMS, SmartLab (SmartLab), Live Engagement, Teams, Chat, Meeting Server, ICLM (Self-Paced & Public Courses), and Public Learning (Learn) modules.

This document provides a comprehensive inventory of all features implemented across the UNILIS system, organized by module. It serves as both a technical reference and a project planning tool, documenting the current implementation status of **95** features across **8** modules.

System Overview

Metric	Value
Total Features	95
Fully Implemented	80 (84%)
Partially Implemented	8
Not Started	7
Modules	8
Version	2.1.0
Last Updated	2026-07-28

2. Feature Summary & Statistics

Implementation Status by Module

Module	Total	Implemented	Partial	Not Started	Completion
Traditional LMS	25	18	5	2	82% ??????????
SmartLab Module	18	18	0	0	100% ??????????
Live Engagement Module	12	12	0	0	100% ??????????
Chat Module	5	5	0	0	100% ??????????
Teams Module	12	12	0	0	100% ??????????
Meeting Server (WebRTC Media Server)	7	7	0	0	100% ??????????
ICLM / Public Learning Module (Learn)	10	10	0	0	100% ??????????
Cross-Cutting Features	9	5	3	1	72% ??????????
TOTAL	95	80	8	7	88%

3. Traditional LMS

Core learning management system with student, lecturer, and admin dashboards

Total Features	Implemented	Partial	Not Started
25	18	5	2

3.1 Authentication System ? IMPLEMENTED

Session-based email/password authentication with role-based access control (student/lecturer/admin)

? login.php, logout.php, forgot_password.php, update_password.php, includes/auth.php

Sub-Features:

- ? IMPLEMENTED Password Reset Flow
- ? IMPLEMENTED Session Management
- ? IMPLEMENTED Role-based Redirects
- ? IMPLEMENTED Student Signup [student/signup.php]

3.2 Student Dashboard ? IMPLEMENTED

Overview of enrolled units, announcements, upcoming deadlines, and grades summary

? student/dashboard.php

Sub-Features:

- ? IMPLEMENTED Enrolled Units Display
- ? IMPLEMENTED Upcoming Deadlines
- ? IMPLEMENTED Grades Summary

3.3 Student Assignments ? IMPLEMENTED

View assignment details, submit work (text/file), track submission status

? student/view_assignments.php, includes/assignment_handler.php

Sub-Features:

- ? IMPLEMENTED Assignment Listing with Deadlines

- **? IMPLEMENTED** File Upload Submission
- **? IMPLEMENTED** Text Submission
- **? IMPLEMENTED** Submission Status Tracking
- **? NOT STARTED** Plagiarism Detection
- **? NOT STARTED** Draft Saving
- **? NOT STARTED** Resubmit Before Deadline

3.4 Student Grades **? IMPLEMENTED**

View graded assignments, unit grades, GPA calculation, and detailed feedback

? student/view_grades.php

Sub-Features:

- **? IMPLEMENTED** Assignment Grades
- **? IMPLEMENTED** Unit Grade Aggregation
- **? NOT STARTED** Grade Analytics
- **? NOT STARTED** Grade Appeal Workflow

3.5 Student Attendance Tracking **? PARTIAL**

View attendance records per unit, mark attendance via QR/Rfid/biometric/code

? student/attendance.php, student/attendance_submit.php, smart-lab/controllers/QrAuthController.php

Sub-Features:

- **? IMPLEMENTED** Attendance Record View
- **? IMPLEMENTED** QR Code Attendance
- **? IMPLEMENTED** RFID Card Attendance
- **? IMPLEMENTED** Biometric Attendance
- **? IMPLEMENTED** Manual Code Entry
- **? NOT STARTED** Attendance Analytics
- **? NOT STARTED** Exemption Requests

3.6 Course Content Access **? IMPLEMENTED**

Access course notes, lecture recordings, slides, and learning materials with lesson view

? student/view_content.php, student/lesson_view.php, student/unit_notes.php, student/js/lesson-dashboard.js

3.7 Student Notifications ? PARTIAL

View announcements, assignment reminders, grade notifications, meeting invitations

? student/notifications.php, includes/notifications.php

Sub-Features:

- ? IMPLEMENTED In-app Notifications
- ? IMPLEMENTED Email Notifications
- ? NOT STARTED Real-time WebSocket Notifications
- ? NOT STARTED Notification Preferences

3.8 Student Profile Management ? IMPLEMENTED

Update personal info, change password, manage notification preferences

? profile.php

3.9 Enrolled Units Query & Management ? IMPLEMENTED

API endpoint to fetch enrolled units for a student with filtering and search

? student/enrolled_units_query.php

3.10 Lecturer Dashboard ? IMPLEMENTED

Overview of teaching load, recent submissions, upcoming assessments, class announcements

? lecturer/dashboard.php, lecturer/lecturer_dashboard.php

3.11 Course & Unit Management ? IMPLEMENTED

Create/edit courses, add units, set schedules, enroll students, manage modules with AJAX save

? lecturer/course_builder.php, lecturer/course_outline.php, lecturer/module_manager.php, lecturer/ajax/save_module.php

Sub-Features:

- ? IMPLEMENTED Course Creation/Editing
- ? IMPLEMENTED Unit Management
- ? IMPLEMENTED Student Enrollment
- ? IMPLEMENTED AJAX Module Saving
- ? NOT STARTED Bulk CSV Enrollment

3.12 Course Content Management ? IMPLEMENTED

Upload lecture slides, recordings, notes; organize by week/topic

? lecturer/coursenotes.php, lecturer/saveClassnotes.php, lecturer/lesson_editor.php

Sub-Features:

- ? IMPLEMENTED File Upload
- ? IMPLEMENTED Content Organization
- ? NOT STARTED Drag-drop Reordering
- ? NOT STARTED Folder Organization

3.13 Assignment Creation & Management ? IMPLEMENTED

Create assignments with rubric, instructions, deadlines; configure grading settings

? lecturer/create_assignment.php, lecturer/assessment_builder.php

Sub-Features:

- ? IMPLEMENTED Assignment Creation with Rubric
- ? IMPLEMENTED Deadline Configuration
- ? IMPLEMENTED Submission Type Configuration
- ? IMPLEMENTED AI Grading Templates
- ? NOT STARTED Reusable Assignment Templates

3.14 Assignment Grading ? IMPLEMENTED

View submissions, enter grades, provide feedback, AI-assisted grading, batch grading

? lecturer/grade_submission.php, lecturer/review_assignments.php, lecturer/submissions.php, includes/ai_grading.php

Sub-Features:

- ? IMPLEMENTED Submission List View
- ? IMPLEMENTED Grade Entry & Feedback
- ? IMPLEMENTED AI-Assisted Grading
- ? IMPLEMENTED Batch Grading
- ? NOT STARTED Rubric-based Grading UI
- ? NOT STARTED Grade Appeal Workflow

3.15 Online Meetings (WebRTC + Meeting Server) ? PARTIAL

Schedule meetings, start WebRTC sessions, record meetings, share screen, live coding IDE, guest access, dedicated Python media server

```
? lecturer/meetings.php, lecturer/meeting_host.php, lecturer/meeting_ide.php, config/meeting.php, api/check_live_meetings.php, meeting_guest.php, meeting_ide.php
```

Sub-Features:

- **? IMPLEMENTED** Meeting Scheduling
- **? IMPLEMENTED** WebRTC Session Hosting
- **? IMPLEMENTED** Meeting Recordings
- **? IMPLEMENTED** Screen Sharing
- **? IMPLEMENTED** Live Coding IDE
- **? IMPLEMENTED** Guest Join (No Login)
- **? IMPLEMENTED** Student Meeting Join [student/meeting_join.php]

3.16 Attendance Management **? PARTIAL**

View unit attendance reports, export records, set exemptions

```
? lecturer/lecturer_take_attendance.php, lecturer/lecturer_attendance_report.php, lecturer/attendance_functions.php
```

Sub-Features:

- **? IMPLEMENTED** Attendance Reports
- **? IMPLEMENTED** Attendance Export
- **? NOT STARTED** Advanced Analytics
- **? NOT STARTED** Exemption Workflow

3.17 File Requests **? IMPLEMENTED**

Request student files for grading/archival with deadline tracking

```
? lecturer/request_files.php
```

Sub-Features:

- **? IMPLEMENTED** File Request Creation
- **? IMPLEMENTED** Deadline Tracking
- **? IMPLEMENTED** Submission Tracking

3.18 Exam & CAT Management **? IMPLEMENTED**

Create exams/CATs, generate questions, manage exam reports, export scores, generate assessment PDFs

? lecturer/create_cat.php, lecturer/create_questions.php, lecturer/exam_reports.php, lecturer/export_scores.php, lecturer/assgenpdf.php

3.19 Course Catalogue Builder ? IMPLEMENTED

Build and manage course catalogues with custom layouts for public display

? lecturer/catalogue_builder.php, lecturer/catalogue_layout.php, lecturer/catalogue.php

3.20 Lecturer Practicals Management ? IMPLEMENTED

Create and manage lab practicals via SmartLab integration (view, grade, monitor)

? lecturer/dashboard.php, smart-lab/controllers/PracticalController.php, smart-lab/views/practicals/index.php

Sub-Features:

- ? IMPLEMENTED Practical Creation
- ? IMPLEMENTED Student Enrollment View
- ? IMPLEMENTED Grading Queue

3.21 Admin Dashboard ? IMPLEMENTED

System overview with user counts, active courses, system health, pending requests

? admin/dashboard.php

3.22 User Management ? PARTIAL

Add/edit users, manage roles, enable/disable accounts, reset passwords

? admin/add_lecturer.php, admin/profile.php, admin/update_password.php

Sub-Features:

- ? IMPLEMENTED User CRUD
- ? IMPLEMENTED Role Management
- ? IMPLEMENTED Admin Profile (Standard/Old Views) [admin/profile_standard.php, admin/profile_old.php]
- ? NOT STARTED Bulk User Import
- ? NOT STARTED Email Verification

3.23 Institution Management ? IMPLEMENTED

Add universities, manage departments, assign department heads

? admin/add_university.php, admin/add_department.php

Sub-Features:

- **? IMPLEMENTED** University CRUD
- **? IMPLEMENTED** Department Management
- **? IMPLEMENTED** Department Head Assignment

3.24 Course & Unit Administration **? IMPLEMENTED**

Add courses, configure units, assign lecturers, manage prerequisites, fetch units via AJAX

? admin/add_course.php, admin/fetch_units.php

3.25 Pending Requests Management **? IMPLEMENTED**

Review and approve/reject team membership requests, file access requests, student appeals

? admin/pendingreq.php

Sub-Features:

- **? IMPLEMENTED** Request Listing
- **? IMPLEMENTED** Approve/Reject Workflow
- **? NOT STARTED** Unified Request System

3.26 Audit Logs **? NOT STARTED**

View system audit logs: user actions, logins, data changes, errors

Sub-Features:

- **? NOT STARTED** Comprehensive Audit Trail
- **? NOT STARTED** Audit Log Viewing

3.27 Admin Reports **? NOT STARTED**

Generate reports: user statistics, course performance, system usage

Sub-Features:

- **? NOT STARTED** Reporting Dashboard
- **? NOT STARTED** Export Capabilities

4. SmartLab Module

Advanced laboratory management system with practicals, notebooks, asset tracking, biometric authentication, and blockchain audit trail

Total Features	Implemented	Partial	Not Started
18	18	0	0

4.1 Multi-Modal Authentication ? IMPLEMENTED

Six authentication methods: password, QR code, RFID, biometric, OTP, and session codes with MFA support

? smart-lab/auth/Auth.php, smart-lab/controllers/AuthController.php, smart-lab/controllers/QrAuthController.php

Sub-Features:

- ? IMPLEMENTED Password Login
- ? IMPLEMENTED QR Code Login (Dynamic)
- ? IMPLEMENTED RFID Card Login
- ? IMPLEMENTED Biometric/Fingerprint Login
- ? IMPLEMENTED OTP Email Login
- ? IMPLEMENTED Session Code Login
- ? IMPLEMENTED Multi-Factor Authentication

4.2 Practical Lifecycle Management ? IMPLEMENTED

Complete practical management from experiment definition to grading with strict academic integrity workflow

? smart-lab/controllers/PracticalController.php, smart-lab/controllers/WorkflowController.php, smart-lab/includes/WorkflowEngine.php

Sub-Features:

- ? IMPLEMENTED Experiment Definition (Technician)
- ? IMPLEMENTED Practical Creation (Lecturer)
- ? IMPLEMENTED Lab Availability Checking
- ? IMPLEMENTED Scheduling & Notification
- ? IMPLEMENTED Student Attendance Verification
- ? IMPLEMENTED Practical Execution Interface

- **? IMPLEMENTED** Report Submission
- **? IMPLEMENTED** Grading & Feedback
- **? IMPLEMENTED** Datasheet Generation

4.3 Academic Integrity Workflow Engine **? IMPLEMENTED**

Strict state machine enforcing physical-presence-only attendance verification with 11 workflow states (Scheduled ? Graded), practical dashboard, and verify attendance views

? smart-lab/includes/WorkflowEngine.php, smart-lab/controllers/WorkflowController.php, smart-lab/migrations/workflow/001_strict_workflow_tables.sql, smart-lab/views/workflow/practical_dashboard.php, smart-lab/views/workflow/verify_attendance.php

Sub-Features:

- **? IMPLEMENTED** 11-State Workflow
- **? IMPLEMENTED** Physical Presence Verification Only
- **? IMPLEMENTED** Verification Window Enforcement
- **? IMPLEMENTED** Automatic Verification Expiry
- **? IMPLEMENTED** Background Checks for Practical Start
- **? IMPLEMENTED** Practical Dashboard for Lecturers
- **? IMPLEMENTED** Attendance Verification Views

4.4 Attendance Verification (Multi-Method) **? IMPLEMENTED**

Mark attendance via Dynamic QR (15-sec refresh), RFID card, biometric fingerprint, technician code, or NFC

? smart-lab/controllers/QrAuthController.php, smart-lab/includes/DynamicQRService.php

Sub-Features:

- **? IMPLEMENTED** Dynamic QR Code (15-sec refresh)
- **? IMPLEMENTED** RFID Card Tap
- **? IMPLEMENTED** Biometric Fingerprint
- **? IMPLEMENTED** Technician Confirmation Code
- **? IMPLEMENTED** NFC Verification
- **? IMPLEMENTED** Nonce-based Anti-replay
- **? IMPLEMENTED** HMAC-SHA256 Digital Signatures

4.5 Datasheet Generation & Verification **? IMPLEMENTED**

Professional JKUAT-branded PDF lab datasheets with QR codes, SHA256 digital signatures, and public verification portal

? smart-lab/includes/DatasheetGenerator.php, smart-lab/includes/DatasheetPDFGenerator.php, smart-lab/includes/DigitalSignature.php, smart-lab/includes/DynamicQRService.php, smart-lab/views/workflow/datasheet_verify.php

Sub-Features:

- ? IMPLEMENTED TCPDF PDF Generation
- ? IMPLEMENTED SHA256 Digital Signatures
- ? IMPLEMENTED Embedded QR Codes
- ? IMPLEMENTED Public Verification Portal
- ? IMPLEMENTED Auto-Approval for Physical Methods
- ? IMPLEMENTED Manual Approval for Password Auth
- ? IMPLEMENTED 365-day QR Token Expiry

4.6 Blockchain Audit Trail ? IMPLEMENTED

Immutable blockchain-based audit trail for datasheet submissions (BlockchainAuditService) and legacy asset tracking (proof-of-work)

? smart-lab/includes/BlockchainAuditService.php, smart-lab/blockchain/Block.php, smart-lab/blockchain/Blockchain.php, smart-lab/blockchain/AssetTracker.php

Sub-Features:

- ? IMPLEMENTED Datasheet Blockchain (SHA-256 chained)
- ? IMPLEMENTED Legacy Asset Blockchain (PoW)
- ? IMPLEMENTED Chain Integrity Validation
- ? IMPLEMENTED Tamper Detection
- ? IMPLEMENTED Asset Transaction History
- ? IMPLEMENTED Full Audit Trail with Provenance

4.7 Digital Lab Notebook ? IMPLEMENTED

Digital lab notebook system with versioning, approval workflow, group sync, and digital signatures on approval

? smart-lab/controllers/NotebookController.php, smart-lab/models/NotebookModel.php

Sub-Features:

- ? IMPLEMENTED Rich Content Notebooks
- ? IMPLEMENTED Version History Tracking
- ? IMPLEMENTED Auto-save
- ? IMPLEMENTED Technician Approval Workflow
- ? IMPLEMENTED Group Notebook Sync

- **? IMPLEMENTED** Digital Signature on Approval

4.8 Asset Management **? IMPLEMENTED**

Laboratory asset inventory management with blockchain-based immutable audit trail (issue/return/transfer/dispose)

? smart-lab/controllers/AssetController.php, smart-lab/blockchain/AssetTracker.php

Sub-Features:

- **? IMPLEMENTED** Asset CRUD
- **? IMPLEMENTED** Asset Lifecycle (issue/return/transfer/dispose)
- **? IMPLEMENTED** Blockchain Tracking
- **? IMPLEMENTED** Lab-to-Lab Transfers
- **? IMPLEMENTED** Usage Logging with Duration
- **? IMPLEMENTED** Min Quantity Alerts

4.9 Practical Request System **? IMPLEMENTED**

Students can request enrollment in practicals they missed or need access to, with admin approval workflow

? smart-lab/controllers/PracticalRequestController.php, smart-lab/controllers/AdminPracticalRequestController.php

Sub-Features:

- **? IMPLEMENTED** Request Submission with Reason
- **? IMPLEMENTED** Status Tracking (pending/approved/rejected)
- **? IMPLEMENTED** Duplicate Prevention
- **? IMPLEMENTED** Activity Logging

4.10 Lab Scheduling & Calendar **? IMPLEMENTED**

Multi-view lab scheduling system with today/week/month/all views, lab filter, occupancy tracking, auth code generation

? smart-lab/controllers/ScheduleController.php

Sub-Features:

- **? IMPLEMENTED** Today/Week/Month/All Views
- **? IMPLEMENTED** Lab Filter
- **? IMPLEMENTED** Schedule Statistics
- **? IMPLEMENTED** Lab Occupancy Tracking

- **? IMPLEMENTED** Auth Code Generation & Email

4.11 Inventory & Lab Monitoring **? IMPLEMENTED**

Lab inventory oversight and CO2 sensor monitoring via Python WebSocket server with real-time dashboard widget

? smart-lab/controllers/InventoryController.php, smart-lab/web-app/smart-lab-server.py, smart-lab/components/co2_monitor_widget.php

Sub-Features:

- **? IMPLEMENTED** Inventory Listing with Statistics
- **? IMPLEMENTED** CO2 Sensor Data Collection
- **? IMPLEMENTED** Real-time CO2 Monitoring Widget
- **? IMPLEMENTED** Lab Occupancy Dashboard

4.12 Role-Based Dashboards **? IMPLEMENTED**

Custom dashboards per user role (student, lecturer, technician, admin) with relevant metrics, system health, and blockchain validation status

? smart-lab/controllers/DashboardController.php, smart-lab/models/DashboardModel.php

Sub-Features:

- **? IMPLEMENTED** Student Dashboard
- **? IMPLEMENTED** Lecturer Dashboard
- **? IMPLEMENTED** Technician Dashboard
- **? IMPLEMENTED** Admin Dashboard
- **? IMPLEMENTED** System Health Indicators
- **? IMPLEMENTED** Blockchain Validation Status
- **? IMPLEMENTED** Weekly Activity Chart
- **? IMPLEMENTED** Lab Utilization Data

4.13 Audit & Logging **? IMPLEMENTED**

Comprehensive activity tracking across all system modules with 30+ action types logged to audit_logs table

? smart-lab/controllers/AuditController.php, smart-lab/models/AuditModel.php

Sub-Features:

- **? IMPLEMENTED** Activity Logging (30+ action types)
- **? IMPLEMENTED** Module-based Logging

- **? IMPLEMENTED** Recent Activity Display
- **? IMPLEMENTED** Weekly Activity Aggregation
- **? IMPLEMENTED** IP Address & User Agent Logging

4.14 Experiment Management **? IMPLEMENTED**

Structured experiment creation by technicians with sections, apparatus list, step-by-step procedures, and results table structure

? smart-lab/controllers/ExperimentController.php

Sub-Features:

- **? IMPLEMENTED** Experiment Definition
- **? IMPLEMENTED** Section-based Content
- **? IMPLEMENTED** Apparatus/Equipment Lists
- **? IMPLEMENTED** Step-by-Step Procedures
- **? IMPLEMENTED** Results Table Structure
- **? IMPLEMENTED** Draft/Publish Workflow

4.15 Report Submission & Grading **? IMPLEMENTED**

Student report submission pipeline with draft/final workflow, file upload, grading with feedback, revision requests, and graded-by tracking

? smart-lab/controllers/ReportController.php, smart-lab/controllers/ReportSubmissionController.php, smart-lab/views/report-submission/, smart-lab/migrations/add_report_graded_by_column.php

Sub-Features:

- **? IMPLEMENTED** Report Creation (Draft/Submitted)
- **? IMPLEMENTED** File Upload (PDF/DOC/DOCX)
- **? IMPLEMENTED** Deadline Tracking with Extensions
- **? IMPLEMENTED** Grading (0-100) with Feedback
- **? IMPLEMENTED** Revision Request Workflow
- **? IMPLEMENTED** Grading Statistics (avg/highest/lowest)
- **? IMPLEMENTED** Report Graded-By Tracking
- **? IMPLEMENTED** Report Submissions Modal View [smart-lab/views/practicals/report_submissions_modal.php]

4.16 User Management (SmartLab) **? IMPLEMENTED**

Role-based user system with registration, staff management, lab assignment, and device fingerprint tracking

? smart-lab/controllers/UsersController.php, smart-lab/auth/Auth.php, smart-lab/create_simple_admin.php

Sub-Features:

- **? IMPLEMENTED** Student Registration
- **? IMPLEMENTED** Staff Registration (with secure key)
- **? IMPLEMENTED** Role-based Lab Assignment
- **? IMPLEMENTED** Account Active/Inactive Status
- **? IMPLEMENTED** Device Fingerprint Tracking
- **? IMPLEMENTED** 5 User Roles (student/lecturer/technician/admin/lab_admin)

4.17 Student Practical Execution **? IMPLEMENTED**

Student-facing practical execution with tabbed interface, auto-save, structured data entry, submission, and test practical support

? smart-lab/controllers/StudentPracticalController.php, smart-lab/controllers/StudentTestPracticalController.php, smart-lab/views/student/view_practical.php, smart-lab/views/student/view_practical_modal_test.php, smart-lab/views/student/test_take_practical.php, smart-lab/views/practicals/index.php

Sub-Features:

- **? IMPLEMENTED** Tabbed Practical Interface
- **? IMPLEMENTED** Auto-save per Section
- **? IMPLEMENTED** Results Table with Dynamic Columns
- **? IMPLEMENTED** Access Window Enforcement
- **? IMPLEMENTED** Draft/Submitted Status Workflow
- **? IMPLEMENTED** Test Practical Mode
- **? IMPLEMENTED** Modal-based Practical View
- **? IMPLEMENTED** Practical Button Check [smart-lab/check_practical_button.php]

4.18 Chemical Datasheet Management **? IMPLEMENTED**

Chemical safety datasheet management with search, PDF generation, and download capabilities

? smart-lab/controllers/DatasheetController.php, smart-lab/models/DatasheetModel.php, smart-lab/views/datasheets.php, smart-lab/api/datasheet.php, smart-lab/api/download_datasheet.php

4.19 Web Routing System **? IMPLEMENTED**

MVC-style web routing with clean URLs, controller dispatch, and .htaccess rewrite rules

? smart-lab/routes/web.php, smart-lab/.htaccess

Sub-Features:

- **? IMPLEMENTED** Route Definitions
- **? IMPLEMENTED** Controller Dispatch
- **? IMPLEMENTED** URL Rewriting (.htaccess)

4.20 Docker & CI/CD Deployment **? IMPLEMENTED**

Containerized deployment with Docker Compose (Apache/PHP + MySQL), GitHub Actions CI/CD pipeline, and production configuration

```
? smart-lab/docker-compose.yml, smart-lab/Dockerfile, smart-lab/apache.conf,  
.github/workflows/main.yml, docker-compose.yml, dockerfile
```

Sub-Features:

- **? IMPLEMENTED** Docker Compose Multi-Service Setup
- **? IMPLEMENTED** GitHub Actions Automated Deployment
- **? IMPLEMENTED** Production Environment Configuration
- **? IMPLEMENTED** Database Initialization Scripts
- **? IMPLEMENTED** Environment Auto-Detection

5. Live Engagement Module

Interactive live engagement module for presentations, polls, quizzes, whiteboards, word clouds, and real-time student participation

Total Features	Implemented	Partial	Not Started
12	12	0	0

5.1 Landing & Home Pages ? IMPLEMENTED

Public landing page and authenticated home dashboard for live engagement features

? modules/live-engagement/index.php, modules/live-engagement/landing.php, modules/live-engagement/views/home.php

Sub-Features:

- ? IMPLEMENTED Public Landing Page
- ? IMPLEMENTED Authenticated Home Dashboard
- ? IMPLEMENTED Setup Script [setup_live_engagement.php]
- ? IMPLEMENTED Bootstrap Initialization [modules/live-engagement/bootstrap.php]

5.2 Presentation Management ? IMPLEMENTED

Create, edit, and deliver interactive presentations with slide support (PDF, PPTX, images, video, HTML)

? modules/live-engagement/views/create_presentation.php, modules/live-engagement/views/presentations.php, modules/live-engagement/views/edit_presentation.php, modules/live-engagement/models/PresentationModel.php

Sub-Features:

- ? IMPLEMENTED Slide Upload (PDF/PPTX/Images/Video/HTML)
- ? IMPLEMENTED Presentation Creation/Editing
- ? IMPLEMENTED Slide Navigation
- ? IMPLEMENTED Up to 200 Slides, 50MB Max

5.3 Live Polls ? IMPLEMENTED

Create and manage live polls with multiple choice (up to 26 options), rating (1-10), and Likert scale types

? modules/live-engagement/api/poll.php, modules/live-engagement/models/PollModel.php

Sub-Features:

- **? IMPLEMENTED** Multiple Choice Polls
- **? IMPLEMENTED** Rating Polls (1-10)
- **? IMPLEMENTED** Likert Scale Polls
- **? IMPLEMENTED** Time-limited Polls (default 120s)
- **? IMPLEMENTED** Real-time Results

5.4 Live Quizzes **? IMPLEMENTED**

Create and conduct live quizzes with time limits, multiple attempts (up to 3), scoring, and up to 100 questions

```
? modules/live-engagement/api/quiz.php, modules/live-engagement/models/QuizModel.php
```

Sub-Features:

- **? IMPLEMENTED** Quiz Creation
- **? IMPLEMENTED** Time Limits (up to 120 min)
- **? IMPLEMENTED** Multiple Attempts (up to 3)
- **? IMPLEMENTED** Passing Score Configuration (default 50%)
- **? IMPLEMENTED** Up to 100 Questions

5.5 Interactive Whiteboard **? IMPLEMENTED**

Real-time collaborative whiteboard with drawing tools, multiple stroke colors/widths, and text tool (up to 5000 objects)

```
? modules/live-engagement/assets/js/live-engagement.js, modules/live-engagement/config/module.php
```

Sub-Features:

- **? IMPLEMENTED** Drawing Tools
- **? IMPLEMENTED** 8 Stroke Colors
- **? IMPLEMENTED** 6 Variable Stroke Widths
- **? IMPLEMENTED** 8 Font Sizes for Text Tool
- **? IMPLEMENTED** Up to 5000 Objects

5.6 Word Cloud **? IMPLEMENTED**

Generate word clouds from participant responses with configurable stop words and up to 100 words

```
? modules/live-engagement/models/WordCloudModel.php
```

Sub-Features:

- **? IMPLEMENTED** Word Collection
- **? IMPLEMENTED** Stop Word Filtering (40+ defaults)
- **? IMPLEMENTED** Up to 100 Words
- **? IMPLEMENTED** Min/Max Word Length Filtering

5.7 Live Session Management **? IMPLEMENTED**

Create and manage live engagement sessions with 8-char session codes, participant join, and auto-close on inactivity

? modules/live-engagement/views/session.php, modules/live-engagement/api/session.php, modules/live-engagement/models/SessionModel.php

Sub-Features:

- **? IMPLEMENTED** Session Creation with 8-Char Codes
- **? IMPLEMENTED** Participant Join Flow
- **? IMPLEMENTED** Session Duration Control (up to 480 min)
- **? IMPLEMENTED** Auto-close on 5-min Inactivity

5.8 Guest Authentication **? IMPLEMENTED**

Allow guest users to join live sessions without full account registration

? modules/live-engagement/api/guest_auth.php

5.9 Premium Design System **? IMPLEMENTED**

Modern glassmorphism design system with dark/light/high-contrast modes, WCAG AA compliance, and animated transitions

? modules/live-engagement/config/module.php, modules/live-engagement/assets/css/live-engagement.css, modules/live-engagement/components/UI.php

Sub-Features:

- **? IMPLEMENTED** Dark Mode
- **? IMPLEMENTED** Light Mode
- **? IMPLEMENTED** High Contrast Mode
- **? IMPLEMENTED** Glassmorphism Effects
- **? IMPLEMENTED** WCAG AA Compliance
- **? IMPLEMENTED** Keyboard Navigation
- **? IMPLEMENTED** Screen Reader Support
- **? IMPLEMENTED** Animated Transitions & Hover Effects

- **? IMPLEMENTED** Ripple Effect & Loading Skeletons
- **? IMPLEMENTED** Material Symbols Rounded Icons

5.10 Real-time Engagement Features **? IMPLEMENTED**

Real-time polling, rate limiting, and reconnection handling for live sessions

? modules/live-engagement/config/module.php, modules/live-engagement/assets/js/live-engagement.js

Sub-Features:

- **? IMPLEMENTED** 3-second Polling Interval
- **? IMPLEMENTED** Rate Limiting (polls/responses/reactions)
- **? IMPLEMENTED** Auto-reconnect (up to 5 attempts)
- **? IMPLEMENTED** CSRF Token Protection
- **? IMPLEMENTED** Activity API Endpoint [modules/live-engagement/api/activity.php]

5.11 Database Management **? IMPLEMENTED**

Database installation scripts, update migrations, and helper utilities for the Live Engagement module

? modules/live-engagement/database/install.php, modules/live-engagement/database/update.php, modules/live-engagement/config/database_helper.php

Sub-Features:

- **? IMPLEMENTED** Install Script
- **? IMPLEMENTED** Update/Migration Script
- **? IMPLEMENTED** Database Helper Utilities

5.12 Security Helpers **? IMPLEMENTED**

Security utilities including session management helpers and security helpers

? modules/live-engagement/helpers/security_helper.php, modules/live-engagement/helpers/session_helper.php, modules/live-engagement/models/BaseModel.php

Sub-Features:

- **? IMPLEMENTED** Security Helper Functions
- **? IMPLEMENTED** Session Helper Functions
- **? IMPLEMENTED** Base Model Class

5.13 Session Join Interface **? IMPLEMENTED**

Join interface for participants to enter live sessions via session code

? modules/live-engagement/views/join.php

5.14 Live Engagement Dashboard ? IMPLEMENTED

Dashboard view for managing and monitoring live engagement sessions

? modules/live-engagement/views/dashboard.php

6. Chat Module

Real-time messaging system for students, lecturers, and staff with conversations, file sharing, groups, and instruction posting

Total Features	Implemented	Partial	Not Started
5	5	0	0

6.1 Core Messaging ? IMPLEMENTED

Full-featured messaging system with conversations, direct messaging, file attachments, and real-time unread counts

? chat/views/chat.php, chat/config.php, chat/api/messages.php, chat/api/send.php, chat/api/conversations.php, chat/api/unread_count.php

Sub-Features:

- ? IMPLEMENTED Conversation Listing
- ? IMPLEMENTED Message Sending & Receiving
- ? IMPLEMENTED Direct Message Start [chat/api/start_direct.php]
- ? IMPLEMENTED Unread Message Counts
- ? IMPLEMENTED Real-time Polling Updates
- ? IMPLEMENTED Deep Linking from Notifications
- ? IMPLEMENTED User Directory Lookup [chat/api/directory.php]

6.2 File Sharing ? IMPLEMENTED

File attachments in messages with upload, serving, and download capabilities

? chat/api/file.php, chat/api/upload.php, chat/includes/chat_files.php

Sub-Features:

- ? IMPLEMENTED File Upload in Messages
- ? IMPLEMENTED File Serving/Download
- ? IMPLEMENTED File Type Validation

6.3 Group Chat & Access Control ? IMPLEMENTED

Group-based chat with access control rules, user badges, and schema management

? chat/includes/chat_access.php, chat/includes/chat_groups.php, chat/includes/chat_repository.php, chat/includes/schema.php

Sub-Features:

- ? IMPLEMENTED Group Chat Support
- ? IMPLEMENTED Access Control Rules
- ? IMPLEMENTED Navigation Badge [chat/includes/nav_badge.php]
- ? IMPLEMENTED Database Schema Setup

6.4 Chat Notifications ? IMPLEMENTED

Notification integration for chat messages, including email notifications

? chat/includes/chat_notify.php

Sub-Features:

- ? IMPLEMENTED Message Notifications
- ? IMPLEMENTED Email Alerts

6.5 Lecturer Instruction Posting ? IMPLEMENTED

Lecturers can post instructions to specific targets (students/groups) with instruction target management

? chat/api/post_instruction.php, chat/api/instruction_targets.php

Sub-Features:

- ? IMPLEMENTED Instruction Posting
- ? IMPLEMENTED Target Selection (Students/Groups)
- ? IMPLEMENTED Instruction History

7. Teams Module

Team management and collaboration features for group projects, submissions, Kanban tasks, standups, checklists, and peer evaluations

Total Features	Implemented	Partial	Not Started
12	12	0	0

7.1 Team Creation & Management ? IMPLEMENTED

Create and manage student teams with invitations, membership requests, approvals, and role management

```
? teams/api/create.php, teams/api/delete_team.php, teams/api/get_team.php,
teams/api/get_user_teams.php, teams/api/get_teams_by_unit.php, teams/api/update_team_settings.php,
teams/api/fetch_teams.php, teams/controllers/TeamController.php, teams/models/Team.php
```

Sub-Features:

- ? IMPLEMENTED Team Creation
- ? IMPLEMENTED Team Settings Updates
- ? IMPLEMENTED Team Deletion
- ? IMPLEMENTED Team Listing/Fetching
- ? IMPLEMENTED Units Assignment [teams/api/get_enrolled_units.php]
- ? IMPLEMENTED Limit Enforcement [teams/includes/team_limits.php]

7.2 Team Membership Management ? IMPLEMENTED

Manage team members with invitations, requests, approvals, removals, and role migration

```
? teams/api/add_member.php, teams/api/remove_member.php, teams/api/get_team_members.php,
teams/api/confirm_member.php, teams/api/request_member_removal.php,
teams/api/request_membership_leave.php, teams/api/accept_invitation.php, teams/api/get_invitations.php,
teams/api/get_team_invitations.php, teams/api/resend_invitation_code.php,
teams/api/cleanup_expired_invitations.php, teams/api/approve_membership_request.php,
teams/api/get_pending_membership_requests.php, teams/api/migrate_membership_requests_table.php,
teams/api/migrate_team_members_role.php, teams/controllers/MemberController.php,
teams/models/TeamMember.php, teams/includes/team_membership.php, teams/includes/ensure_team_marks.php
```

Sub-Features:

- ? IMPLEMENTED Add/Remove Members
- ? IMPLEMENTED Invitation System with Codes
- ? IMPLEMENTED Membership Requests & Approval

- **? IMPLEMENTED** Membership Leave/Removal Requests
- **? IMPLEMENTED** Expired Invitation Cleanup
- **? IMPLEMENTED** Role Migration
- **? IMPLEMENTED** Team Marks Tracking

7.3 Team Views & UI **? IMPLEMENTED**

User-facing views for team management including workspace, create team, manage team, approve requests

? teams/views/workspace.php, teams/views/create_team.php, teams/views/manage_team.php, teams/views/approve_membership_requests.php, teams/views/lecturer_teams.php

Sub-Features:

- **? IMPLEMENTED** Team Workspace (Full UI)
- **? IMPLEMENTED** Team Creation Form
- **? IMPLEMENTED** Team Management UI
- **? IMPLEMENTED** Membership Request Approval UI
- **? IMPLEMENTED** Lecturer Team Overview

7.4 Team Submissions & Assessments **? IMPLEMENTED**

Submit assignments as a team, lecturer assessment of team submissions, file management, download submissions

? teams/api/submit.php, teams/api/get_submissions.php, teams/api/download_submission.php, teams/api/lecturer_assessments.php, teams/api/get_assessments.php, teams/api/award_mark.php, teams/api/delete_team_file.php, teams/api/view_team_file.php, teams/controllers/SubmissionController.php, teams/models/TeamSubmission.php, teams/views/submit.php

Sub-Features:

- **? IMPLEMENTED** Team Assignment Submission
- **? IMPLEMENTED** Lecturer Assessment & Marking
- **? IMPLEMENTED** Submission Download
- **? IMPLEMENTED** Team File Management
- **? IMPLEMENTED** Assessment Listings

7.5 Kanban Task Board **? IMPLEMENTED**

Kanban-style task management for teams with task creation, status updates, and board view

? teams/api/kanban_tasks.php, teams/api/task_create.php, teams/api/task_update_status.php

Sub-Features:

- **? IMPLEMENTED** Task Creation
- **? IMPLEMENTED** Task Status Updates (Drag & Drop)
- **? IMPLEMENTED** Kanban Board API

7.6 Daily Standups **? IMPLEMENTED**

Daily standup tracking for team members with create and view standups

? teams/api/standups.php, teams/api/standup_create.php

Sub-Features:

- **? IMPLEMENTED** Standup Creation
- **? IMPLEMENTED** Standup History Viewing

7.7 Team Checklists **? IMPLEMENTED**

Checklist management with toggle completion and sign-off workflow

? teams/api/checklist.php, teams/api/checklist_toggle.php, teams/api/checklist_signoff.php

Sub-Features:

- **? IMPLEMENTED** Checklist Creation
- **? IMPLEMENTED** Item Toggle (Complete/Incomplete)
- **? IMPLEMENTED** Checklist Sign-off

7.8 Peer Evaluation **? IMPLEMENTED**

Peer evaluation system with submission, summary reports, and lecturer-facing reports

? teams/api/peer_evaluation_submit.php, teams/api/peer_evaluation_summary.php, teams/api/peer_evaluation_report.php

Sub-Features:

- **? IMPLEMENTED** Peer Evaluation Submission
- **? IMPLEMENTED** Peer Evaluation Summary
- **? IMPLEMENTED** Lecturer Evaluation Reports

7.9 Team File Requests **? IMPLEMENTED**

Student and lecturer file request management with submission tracking

? teams/api/student_file_requests.php, teams/api/lecturer_file_requests.php, teams/views/file_requests.php

Sub-Features:

- **? IMPLEMENTED** Student File Requests View
- **? IMPLEMENTED** Lecturer File Requests Management

7.10 Activity Logging **? IMPLEMENTED**

Track team activity with activity log for audit and monitoring

```
? teams/api/get_activity_log.php, teams/models/ActivityLog.php
```

Sub-Features:

- **? IMPLEMENTED** Activity Log Retrieval
- **? IMPLEMENTED** Activity Log Model

7.11 Lecturer Team Insights **? IMPLEMENTED**

Lecturer insights dashboard with team performance data and ghost nudging (inactive member alerts)

```
? teams/api/lecturer_team_insights.php, teams/api/lecturer_units.php, teams/api/ghost_nudge.php, teams/api/ghost_status.php, teams/api/health.php
```

Sub-Features:

- **? IMPLEMENTED** Team Performance Insights
- **? IMPLEMENTED** Unit-based Lecturer View
- **? IMPLEMENTED** Ghost Nudging (Inactive Members)
- **? IMPLEMENTED** Ghost Status Tracking
- **? IMPLEMENTED** Health Check API

7.12 Workspace File Management **? IMPLEMENTED**

Team workspace file storage with serving, listing, and management

```
? teams/api/workspace_files.php, teams/api/serve_file.php
```

Sub-Features:

- **? IMPLEMENTED** Workspace File Listing
- **? IMPLEMENTED** File Serving

7.13 Team Data Export **? IMPLEMENTED**

Export team data in PDF and Excel formats including group members and all teams

? teams/api/export_group_members_pdf.php, teams/api/export_pdf.php, teams/api/export_excel.php,
teams/api/export_all_teams_pdf.php, teams/api/export_all_teams_excel.php

8. Meeting Server (WebRTC Media Server)

Python FastAPI-based WebRTC signaling and collaborative meeting server with real-time chat, polls, whiteboard, attendance tracking, and media compositing

Total Features	Implemented	Partial	Not Started
7	7	0	0

8.1 WebRTC Signaling & Room Management ? IMPLEMENTED

FastAPI-based WebRTC signaling server with room creation, participant management, capability/policy enforcement, and CORS configuration

? meeting-server/server.py, meeting-server/room_manager.py, meeting-server/requirements.txt, meeting-server/Dockerfile

Sub-Features:

- ? IMPLEMENTED Room Create/Join/Leave
- ? IMPLEMENTED Participant Management
- ? IMPLEMENTED Capability & Policy Enforcement
- ? IMPLEMENTED CORS Middleware
- ? IMPLEMENTED Optional Authentication via API Key/Header
- ? IMPLEMENTED OpenAPI Documentation (FastAPI auto)

8.2 WebSocket Manager ? IMPLEMENTED

WebSocket connection management with reconnection handling, broadcast, and user-specific messaging

? meeting-server/websocket_manager.py

Sub-Features:

- ? IMPLEMENTED WebSocket Connection Pool
- ? IMPLEMENTED Broadcast to Room/All
- ? IMPLEMENTED User-specific Messaging
- ? IMPLEMENTED Reconnection Handling

8.3 Meeting Chat ? IMPLEMENTED

Real-time chat within meetings with message history and rate limiting


```
? meeting-server/chat_handler.py
```

Sub-Features:

- **? IMPLEMENTED** Real-time Chat Messages
- **? IMPLEMENTED** Message History
- **? IMPLEMENTED** Rate Limiting

8.4 Meeting Polls **? IMPLEMENTED**

In-meeting polling system with poll creation, voting, and real-time results

```
? meeting-server/poll_handler.py
```

Sub-Features:

- **? IMPLEMENTED** Poll Creation
- **? IMPLEMENTED** Voting System
- **? IMPLEMENTED** Real-time Results Broadcast

8.5 Meeting Whiteboard **? IMPLEMENTED**

Real-time collaborative whiteboard within meetings with drawing tools and state synchronization

```
? meeting-server/whiteboard_handler.py
```

Sub-Features:

- **? IMPLEMENTED** Whiteboard State Sync
- **? IMPLEMENTED** Drawing Tool Support
- **? IMPLEMENTED** Real-time Broadcast

8.6 Meeting Attendance Tracking **? IMPLEMENTED**

Automatic attendance tracking for meeting participants with join/leave timestamps

```
? meeting-server/attendance_handler.py
```

Sub-Features:

- **? IMPLEMENTED** Join/Leave Timestamping
- **? IMPLEMENTED** Duration Tracking
- **? IMPLEMENTED** Attendance Records

8.7 Media Room & Frame Compositing **? IMPLEMENTED**

OpenCV-based camera/screen-share stream compositing with participant frames, JPEG encoding, and composite frame generation

? meeting-server/media_room.py

Sub-Features:

- ? IMPLEMENTED Camera Frame Capture
- ? IMPLEMENTED Screen Share Frame Capture
- ? IMPLEMENTED Composite Frame Generation (OpenCV)
- ? IMPLEMENTED JPEG Encoding (Quality 75)
- ? IMPLEMENTED 1280x720 Default Resolution
- ? IMPLEMENTED 20fps Frame Processing

9. ICLM / Public Learning Module (Learn)

Self-paced online certification courses with public catalogue, enrolment, lessons, assessments, certificates, and verification system - fully backend-driven

Total Features	Implemented	Partial	Not Started
10	10	0	0

9.1 Public Course Catalogue ? IMPLEMENTED

Public-facing course catalogue with search, course cards, hero section, lesson/assessment counts, and learner enrollment tracking

? learn/index.php, learn/config.php

Sub-Features:

- ? IMPLEMENTED Searchable Course Listing
- ? IMPLEMENTED Course Cards with Metadata (Level, Lessons, Certificate)
- ? IMPLEMENTED Learner Count Display
- ? IMPLEMENTED Hero Section
- ? IMPLEMENTED Responsive Grid Layout

9.2 Learner Authentication & Registration ? IMPLEMENTED

Separate authentication system for external learners with registration, login, password reset, email verification, and session management

? learn/login.php, learn/register.php, learn/logout.php, learn/includes/learner_auth.php

Sub-Features:

- ? IMPLEMENTED Learner Registration
- ? IMPLEMENTED Login with Session Management
- ? IMPLEMENTED Email Verification Link (48hr TTL)
- ? IMPLEMENTED Minimum 10-char Password Policy
- ? IMPLEMENTED Separate Session from LMS Users

9.3 Course Page & Learning Interface ? IMPLEMENTED

Full course page with module/lesson outline, enrolment button, lesson reading, completion tracking, and final assessments

? learn/course.php, learn/includes/catalogue.php, learn/includes/layout.php

Sub-Features:

- ? IMPLEMENTED Module-based Course Outline
- ? IMPLEMENTED Lesson Reading Interface
- ? IMPLEMENTED Lesson Completion Marking
- ? IMPLEMENTED Course Enrolment (Self-enrol)
- ? IMPLEMENTED Progress Tracking (lessons + assessments)
- ? IMPLEMENTED CSRF-protected Actions

9.4 Assessments & Grading ? IMPLEMENTED

Server-side graded assessments with pass marks, multiple attempts tracking, and correct answer hiding until submission

? learn/assessment.php, learn/includes/catalogue.php

Sub-Features:

- ? IMPLEMENTED Server-side Grading
- ? IMPLEMENTED Pass Mark Configuration
- ? IMPLEMENTED Multiple Attempts Tracking
- ? IMPLEMENTED Course-level Pass Mark
- ? IMPLEMENTED Enrolment-based Access Control

9.5 Certificate Generation & Verification ? IMPLEMENTED

Verifiable certificates with unique 48-hex-character codes, public verification page, screen/print-ready rendering, and revocation support

? learn/certificate.php, learn/verify_certificate.php, learn/includes/catalogue.php

Sub-Features:

- ? IMPLEMENTED Certificate Generation on Course Completion
- ? IMPLEMENTED Unique 48-hex Verification Codes
- ? IMPLEMENTED Public Verification Page (No Login Required)
- ? IMPLEMENTED Print-ready Certificate Layout
- ? IMPLEMENTED Certificate Revocation Support

9.6 Learner Dashboard ? IMPLEMENTED

Personal learning dashboard showing enrolled courses, progress, and certificates

```
? learn/dashboard.php
```

Sub-Features:

- ? IMPLEMENTED Enrolled Courses List
- ? IMPLEMENTED Progress per Course
- ? IMPLEMENTED Certificate Status

9.7 Course Authoring Tools ? IMPLEMENTED

Authoring system for creating and managing public courses, modules, lessons, and assessments with slug-based URLs

```
? learn/includes/authoring.php
```

Sub-Features:

- ? IMPLEMENTED Course CRUD
- ? IMPLEMENTED Module Management
- ? IMPLEMENTED Lesson Content Creation
- ? IMPLEMENTED Assessment Creation with Questions
- ? IMPLEMENTED Publish/Unpublished Workflow
- ? IMPLEMENTED Slug-based Course URLs

9.8 Learner Email System ? IMPLEMENTED

Email notifications for learners including verification emails and password resets

```
? learn/includes/learner_mail.php
```

Sub-Features:

- ? IMPLEMENTED Verification Emails
- ? IMPLEMENTED Password Reset Emails

9.9 Standalone Certificate Verification ? IMPLEMENTED

Dedicated verification page where anyone can enter a certificate code to verify authenticity

```
? learn/verify.php, learn/verify_certificate.php
```

9.10 Static Course Catalog (Self-Paced) ? IMPLEMENTED

Legacy static HTML/CSS course catalog page with categories, carousel, and golden theme

? self_paced/onlinecourse.php

Sub-Features:

- ? IMPLEMENTED Course Categories (IT, Business, Design, Marketing, Language)
- ? IMPLEMENTED Featured Courses Carousel
- ? IMPLEMENTED Golden/Cream Theme Design

10. Cross-Cutting Features

Features that span across multiple modules

Total Features	Implemented	Partial	Not Started
9	5	3	1

10.1 Email System ? IMPLEMENTED

PHPMailer-based email system with templates, logging, retry mechanism, deadline reminders, and bulk sending

? includes/mailer.php, includes/email_system.php, includes/deadline_reminders.php, config/email.php

Sub-Features:

- ? IMPLEMENTED Bulk Email Sending
- ? IMPLEMENTED Email Templates (welcome, reset, grades, announcements)
- ? IMPLEMENTED Email Logging (email_logs table)
- ? IMPLEMENTED Retry Mechanism for Failed Sends
- ? IMPLEMENTED Deadline Reminder Emails
- ? IMPLEMENTED Cron-based Deadline Reminders [setup/deadline_reminder_cron.php]
- ? IMPLEMENTED Email Test Utility [setup/test_email_system.php]
- ? NOT STARTED Email Queue System
- ? NOT STARTED Email Preview Functionality

10.2 File Handling System ? PARTIAL

File upload, storage, and management across the system with type/size validation

? assets/uploads/, assets/assignments/, assets/meetings/, assets/requested_files/

Sub-Features:

- ? IMPLEMENTED File Upload with Validation
- ? IMPLEMENTED File Type/Size Restrictions
- ? IMPLEMENTED Base64 File Upload [smart-lab/public/upload_base64.php]
- ? NOT STARTED Upload Outside Webroot
- ? NOT STARTED Virus Scanning
- ? NOT STARTED File Encryption

10.3 Security Measures ? PARTIAL

Security implementations across the system including BCrypt hashing, SQL injection prevention, and XSS prevention

Sub-Features:

- ? IMPLEMENTED Password Hashing (BCRYPT)
- ? IMPLEMENTED SQL Injection Prevention (Prepared Statements)
- ? IMPLEMENTED XSS Prevention (HTML escaping, sanitization)
- ? IMPLEMENTED Session-based Authentication
- ? IMPLEMENTED Role-based Access Control
- ? IMPLEMENTED HMAC-SHA256 QR Token Signing
- ? IMPLEMENTED SHA256 Digital Signatures
- ? IMPLEMENTED CSRF Token Protection
- ? NOT STARTED Rate Limiting (login, API)
- ? NOT STARTED HTTPS Enforcement
- ? NOT STARTED Security Headers (CSP, X-Frame-Options)
- ? NOT STARTED Two-Factor Authentication

10.4 Deployment & DevOps ? IMPLEMENTED

Docker containerization, GitHub Actions CI/CD pipeline, production configuration, and database initialization

? docker-compose.yml, Dockerfile, apache.conf, .github/workflows/main.yml, start.sh, docker/init.sh

Sub-Features:

- ? IMPLEMENTED Docker Compose Multi-Service Setup
- ? IMPLEMENTED GitHub Actions CI/CD Pipeline
- ? IMPLEMENTED Production Configuration Files
- ? IMPLEMENTED Database Initialization Scripts
- ? IMPLEMENTED Environment Auto-Detection (Dev/Prod)
- ? IMPLEMENTED Database Backup Scripts

10.5 Data Validation ? IMPLEMENTED

Client-side and server-side validation framework with sanitization and prepared statements

? includes/validation.php, smart-lab/utils/Validator.php, smart-lab/utils/helpers.php

Sub-Features:

- **? IMPLEMENTED** Required Field Validation
- **? IMPLEMENTED** Email Format Validation
- **? IMPLEMENTED** File Type/Size Validation
- **? IMPLEMENTED** Date/Deadline Validation
- **? IMPLEMENTED** XSS Sanitization (sanitize/sanitizeHTML)
- **? NOT STARTED** Centralized Rules Engine

10.6 Testing **? NOT STARTED**

Automated testing infrastructure not yet implemented across the system

Sub-Features:

- **? NOT STARTED** Unit Tests (PHPUnit)
- **? NOT STARTED** Integration Tests
- **? NOT STARTED** UI Tests (Playwright)
- **? NOT STARTED** API Tests

10.7 Performance Optimization **? NOT STARTED**

Performance optimizations needed across the system including indexing, caching, and asset optimization

Sub-Features:

- **? NOT STARTED** Database Indexing
- **? NOT STARTED** Query Optimization (N+1 prevention)
- **? NOT STARTED** Caching Layer (Redis)
- **? NOT STARTED** Response Compression (gzip)
- **? NOT STARTED** Asset Minification (CSS/JS)
- **? NOT STARTED** Image Optimization
- **? NOT STARTED** Lazy Loading

10.8 Notification System **? PARTIAL**

Multi-channel notification system with in-app and email notifications for various triggers

? includes/notifications.php, student/notifications.php, lecturer/dashboard.php

Sub-Features:

- **? IMPLEMENTED** In-app Notifications (database)
- **? IMPLEMENTED** Email Notifications (PHPMailer)
- **? IMPLEMENTED** Assignment Deadline Reminders

- **? IMPLEMENTED** Grade Released Notifications
- **? IMPLEMENTED** Meeting Invitations
- **? IMPLEMENTED** Chat Message Notifications
- **? IMPLEMENTED** Deadline Reminder Cron
- **? NOT STARTED** WebSocket Real-time Notifications
- **? NOT STARTED** SMS Notifications
- **? NOT STARTED** Push Notifications
- **? NOT STARTED** Notification Preferences UI

10.9 Database Migrations & Fixes **? IMPLEMENTED**

Database migration scripts and data fixes for schema updates across modules

```
? migrations/, fix/, smart-lab/migrations/
```

Sub-Features:

- **? IMPLEMENTED** Attendance Table Migration
- **? IMPLEMENTED** Chat System Migration
- **? IMPLEMENTED** Meeting Guests Migration
- **? IMPLEMENTED** External Learners Migration
- **? IMPLEMENTED** Note Status Migration
- **? IMPLEMENTED** RFID Database Migration
- **? IMPLEMENTED** Database Collation Fix
- **? IMPLEMENTED** SmartLab Workflow Migrations
- **? IMPLEMENTED** Report Graded-By Migration

11. Known Issues & Limitations

- Practical attendance modal QR scanning not fully tested in browser
- Lab session start flow and notebook editor integration need completion
- Biometric enrollment flow not implemented in student profile
- RFID card registration UI not implemented
- WebRTC meeting system partially implemented (screen sharing incomplete)
- Real-time notifications use AJAX polling instead of WebSocket
- Audit logging not comprehensive for traditional LMS
- No automated test suite across the system
- File uploads stored in webroot (security concern)
- API endpoints lack rate limiting and CSRF protection
- Two separate authentication systems (traditional LMS and smart-lab) with different user tables
- Admin reports and audit log viewing not implemented
- ICLM module has separate learner auth from LMS auth systems
- Meeting server requires Python/FastAPI deployment separately from PHP
- Dual authentication/user systems (LMS students, SmartLab users, Learn learners) need consolidation

12. Recommended Next Steps

1. 1. Test practical attendance modal in browser (QR code, RFID, biometric, code)
2. 2. Complete practical session flow (session start, notebook editor integration, grading)
3. 3. Implement biometric and RFID enrollment flows in student profile
4. 4. Complete WebRTC meeting system (screen sharing, recording)
5. 5. Implement real-time WebSocket notifications
6. 6. Create comprehensive test suite (unit, integration, UI, API)
7. 7. Implement security hardening (CSRF, rate limiting, HTTPS, security headers)
8. 8. Optimize database queries and add caching layer (Redis)
9. 9. Implement comprehensive audit logging across all modules
10. 10. Complete admin reporting dashboard with export capabilities
11. 11. Consolidate dual/three authentication systems into unified user management
12. 12. Deploy meeting-server as a standalone Python microservice
13. 13. Set up monitoring, logging, and alerting for production
14. 14. Complete Teams module integration with lecturer dashboards
15. 15. Expand ICLM/Learn module with payment gateway integration